

Phage Anti-Biofilm Coating for Shunt Infection Prevention

Executive Technology Brief

Domain: Microbiology + Surface Science + Biomaterials

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Infection causes 8-12% of shunt failures at \$40-80K per case

Mechanism: S. aureus phage K immobilized on electrospun titanium-coated catheter surface prevents biofilm establishment

Architecture: CSF shunt catheter with S. aureus phage K immobilized on electrospun titanium-coated surface; phages prevent biofilm formation by infecting and lysing S. aureus before biofilm establishes

Governing Model: Phage-bacteria adsorption kinetics on immobilized phage surface. Surface phage density must exceed threshold such that bacterial lysis rate exceeds colonization rate during the critical 24-48h post-implantation window.

Key Equations:

$dP/dt = -k_{ads} * [Phage] * [Bacteria]$ [phage-bacteria binding kinetics]

$\mu = \mu_{max} * S / (K_s + S)$ [Monod specific growth rate]

$\Gamma_{phage}(t) = \Gamma_0 * \exp(-k_{decay} * t)$ [phage surface density decay]

Threshold: $k_{lysis} * \Gamma_{phage}(t) > k_{attach} * [Bact]_{challenge}$ for t in $[0, t_{critical}]$

Kill Condition: Phage K cannot retain infectivity when immobilized on Ti surface for >24h

Current State: ENGINEERING_DEFINITION. No physical prototype. No clinical validation.

Disclosure: This dossier distinguishes established evidence from proposals and unresolved questions.